Technische Hochschule Ingolstadt
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Autonomes Fliegen

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Kapitel

- **Kapitel 1: Übersicht über Unmanned Aerial Systems (UAVs) (1 W)**
- **Kapitel 2: Wesentliche Komponenten/Subsysteme eines UAV (1 W)**
- **Kapitel 3: Modellierung und Simulation von UAVs (1 W)**
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- **Kapitel 5: Pfadplanung (4 W)**
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Nonlinear Filters for Optimal Trajectory Planning

- Idea
  - Instead of precomputing (offline) a full jerk-/acceleration-/velocity-limited trajectory for every change in task or limits, the chapter builds dynamic nonlinear filters that sit after a simple reference generator (steps, ramps) and transform it online into a feasible, near–minimum-time trajectory obeying bounds on $\dot{q}, \ddot{q}, \dddot{q}$. Think "trajectory = reference → smart filter → admissible motion," all in real time.

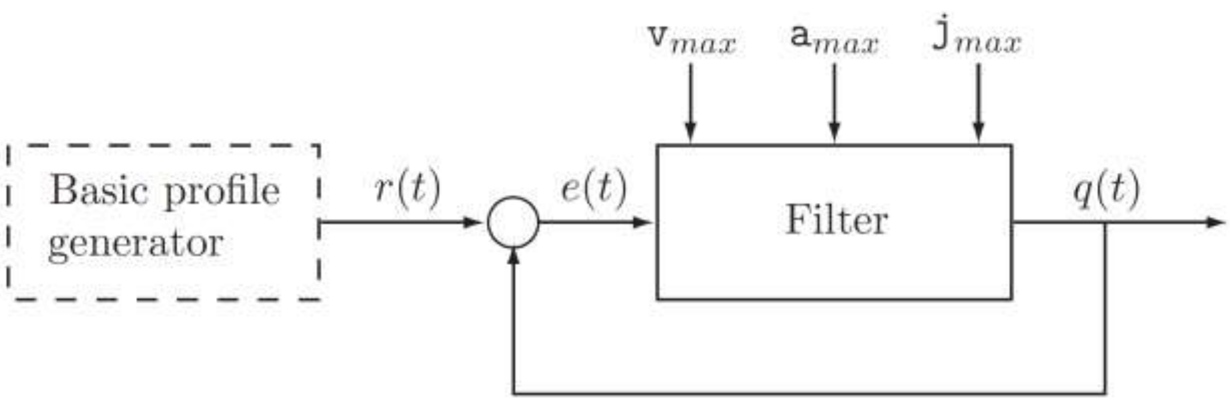

Fig. 1. Conceptual scheme of the nonlinear filter for online optimal trajectory generation.

Architecture at a glance

- A basic profile $r(t)$ step, ramp, time-varying command) feeds a feedback nonlinear filter that outputs $q(t)$ while enforcing bounds; the filter tries to track r as aggressively as possible without violating limits (conceptual scheme in Fig. 1).

  - Constraints enforced (third-order case):

$$\begin{aligned} \mathtt{v}_{min} \leq & \quad \dot{q}(t) \quad \leq \mathtt{v}_{max} \\ \mathtt{a}_{min} \leq & \quad \ddot{q}(t) \quad \leq \mathtt{a}_{max} \\ -U = \mathtt{j}_{min} \leq & \; q^{(3)}(t) \leq \mathtt{j}_{max} = U. \end{aligned}$$

✓ The input may be a live signal (sensor), not only a preplanned profile.

Third-order filter: online planning with jerk limits (double-S segments) 1

- Discrete realization

  - Sampling at $t_k = kT_s$ .The control variable is **jerk** $u_k \triangleq \dddot{q}$. Three integrators map jerk to position (rectangular for $\ddot{q}$;trapezoidal for $\dot{q},$):

$$\ddot{q}_k = \ddot{q}_{k-1} + T_s u_{k-1}$$
$$\dot{q}_k = \ddot{q}_{k-1} + \frac{T_s}{2}(\ddot{q}_k + \ddot{q}_{k-1})$$
$$q_k = q_{k-1} + \frac{T_s}{2}(\dot{q}_k + \ddot{q}_{k-1}).$$

  - Error normalization (with as jerk scale):

$$e_k = \frac{q_k - r_k}{U}, \qquad \dot{e}_k = \frac{\dot{q}_k - \dot{r}_k}{U}, \qquad \ddot{e}_k = \frac{\ddot{q}_k - \ddot{r}_k}{U}$$

Third-order filter: online planning with jerk limits (double-S segments) 2

- Velocity/acceleration bounds become time-varying error bounds because they depend on $\dot{r}_k, \ddot{r}_k$and are recomputed every sample. The variable-structure law C_3)sliding-mode flavored) chooses $\mathrm{u_k} \in \{\pm \mathrm{U}\}$ or a saturated value so that q rushes toward r while staying inside the moving $(\dot{\mathrm{q}}, \ddot{\mathrm{q}})$ admissible box; the resulting motion is piecewise-constant-jerk (i.e., double-S). (See block diagram Fig. 3 and explicit controller expressions.)

- **Key properties in practice**
  - Produces continuous $q, \dot{q}, \ddot{q}$,piecewise-constant $\dddot{q}$)double-S segments)
  - Online editability of limits: change $v_{\min/max}, a_{\min/max}$during motion
  - Tracks time-varying references (e.g., sawtooth or conveyorsynchronization) while obeying limits

Online planning with acceleration limits (trapezoidal velocity)

- Small position/acceleration overshoots and chattering in commanded jerk around zero; these shrink with smaller T_s

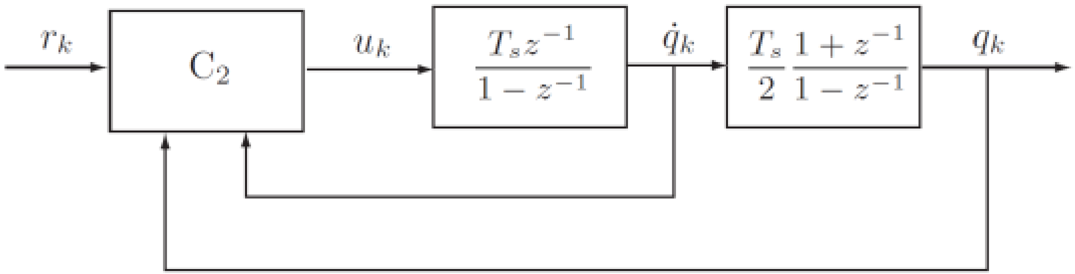

Fig. 2. Second order filter for online optimal trajectory generation.

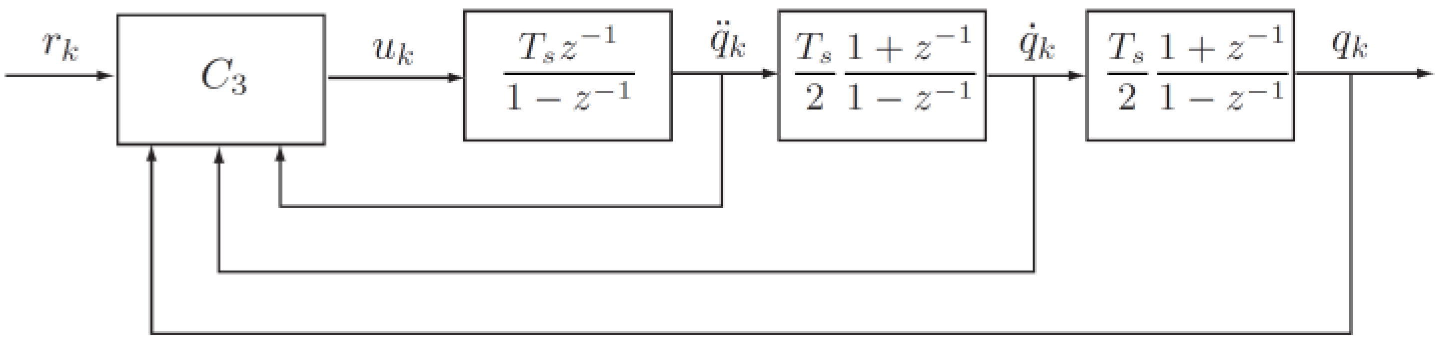

Fig. 3. Third order filter for online optimal trajectory generation.

- The control signal computed as

$$
\begin{cases}
z_k = \dfrac{1}{T_s}\left(\dfrac{e_k}{T_s} + \dfrac{\dot{e}_k}{2}\right), \qquad \dot{z}_k = \dfrac{\dot{e}_k}{T_s} \\
m = \operatorname{floor}\left(\dfrac{1 + \sqrt{1 + 8|z_k|}}{2}\right) \\
\sigma_k = \dot{z}_k + \dfrac{z_k}{m} + \dfrac{m-1}{2}\operatorname{sign}(z_k) \\
u_k = -U \operatorname{sat}(\sigma_k)\, \dfrac{1 + \operatorname{sign}(\dot{q}_k \operatorname{sign}(\sigma_k) + \mathrm{v}_{max} - T_s U)}{2}
\end{cases}
$$

where $\operatorname{floor}(\cdot)$ is the function 'integer part', $\operatorname{sign}(\cdot)$ is the sign function, and $\operatorname{sat}(\cdot)$ is a saturation function defined by

$$
\operatorname{sat}(x) = \begin{cases} -1, & x < -1 \\ x, & -1 \leq x \leq 1 \\ +1, & x > 1. \end{cases}
$$

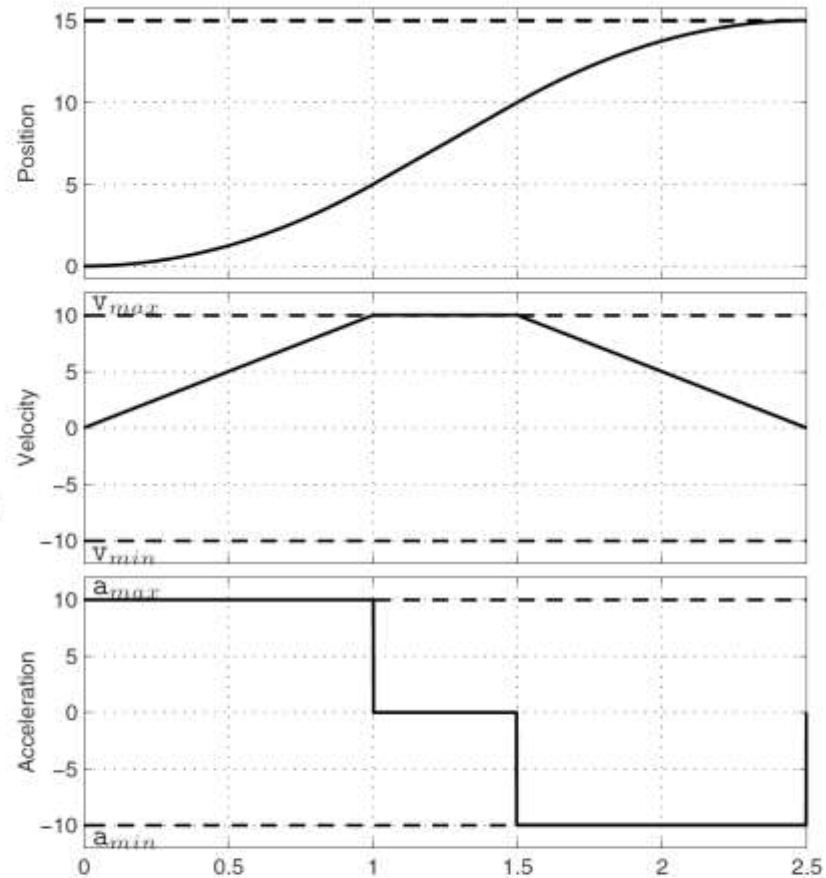

Fig. 4. Output (position, velocity and acceleration) of the second order nonlinear filter with a step reference signal.